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# TD Prediction

Instructor

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# Temporal Difference (TD) Learning to the rescue

- Incremental every-visit Monte-Carlo
  - Update value $v(s_t)$ towards actual return $G_t$

$$v(S_t) \leftarrow v(S_t) + \alpha(G_t - v(S_t))$$

- Recall, the return

$$G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T = R_{t+1} + \gamma v(S_{t+1})$$

- Temporal-Difference Learning Algorithm
  - Basic Idea: Exploit Markovian property to replace actual return $G_t$ with estimated return $R_{t+1} + \gamma v(S_{t+1})$
  - Update value towards estimated return

$$v(s_t) \leftarrow v(s_t) + \alpha \underbrace{\left( \overbrace{R_{t+1} + \gamma v(s_{t+1})}^{\text{TD target}} - v(s_t) \right)}_{\text{TD error}}$$

# Temporal Difference (TD) Learning To The Rescue

$$v(s_t) \leftarrow v(s_t) + \alpha \underbrace{\left( \overbrace{R_{t+1} + \gamma v(s_{t+1})}^{\text{TD target}} - v(s_t) \right)}_{\text{TD error}}$$

# MC vs TD(0)

## Monte-Carlo

Return $G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$
is an unbiased estimate of
$v(s_t)$

Return depends on many random actions, transitions, rewards

Insensitive to initial values

Better convergence properties

Applies to episodic MDPs

## Temporal-Difference

Return $G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$
is a biased estimate of
$v(s_t)$

Return depends on one random actions, transitions, rewards

Sensitive to initial values

May not converge with function approximation

Applies to non-episodic MDPs, too

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# TD (λ) n-Step Return

Why use only next-state for estimating return in TD learning?

Recall: $G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-t-1} R_T$

One may consider an n-step return:

$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n} + \gamma^n v(S_{t+n})$$

$n = 1$ — TD (0)

$n = \infty$ — MC

n-step TD update:

$$v(S_t) \leftarrow v(S_t) + \alpha\left(G_t^{(n)} - v(S_t)\right)$$

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# TD (λ) Averaging n-Step Return

Can we combine information from multiple time-steps?

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# TD (λ) Averaging n-Step Return

Instead of using just 1-step return, 2-step return, etc., we can use generalized return, such as below:

$$G_t = 0.5 \cdot G_t^{(1)} + 0.3 \cdot G_t^{(2)} + 0.2 \cdot G_t^{(3)}$$

How can we combine information from all time-steps?

$$G_t^{\lambda} = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$$

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# TD (λ) Averaging n-Step Return

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Forward-view TD (λ) algorithm:

$$v(S_t) \leftarrow v(S_t) + \alpha \left( G_t^{(\lambda)} - v(S_t) \right)$$

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# Thank You